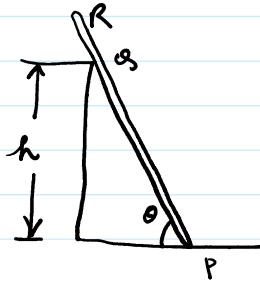
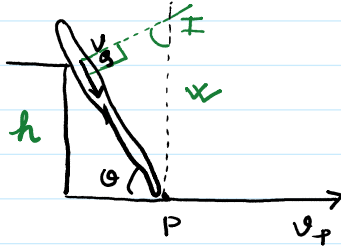


①



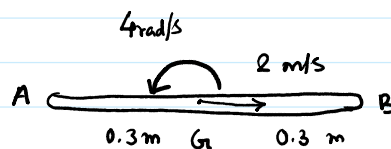
A slender prismatic rod PR has a support at Q on the wall. If point P has velocity v_P in horizontal direction, determine angular velocity of the rod.

Soln: At this instant of time, if I can find out I-centre.

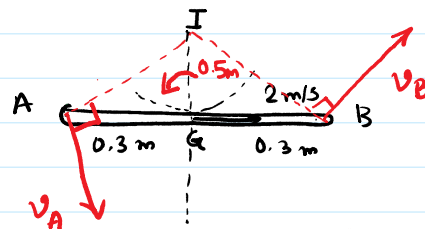


$$\omega = \frac{v_P}{IP} = \frac{v_Q}{IQ}$$

②

Inst

The slender bar is moving in general plane motion with the indicated linear and angular properties. Locate the instantaneous centre of zero velocity and determine velocities of point A and B.

Soln:

4 rad/s

 $\omega_G = 4 \text{ rad/s}$

$$\frac{v_G}{IG} = \omega = 4.$$

$$\frac{2}{IG} = 4$$

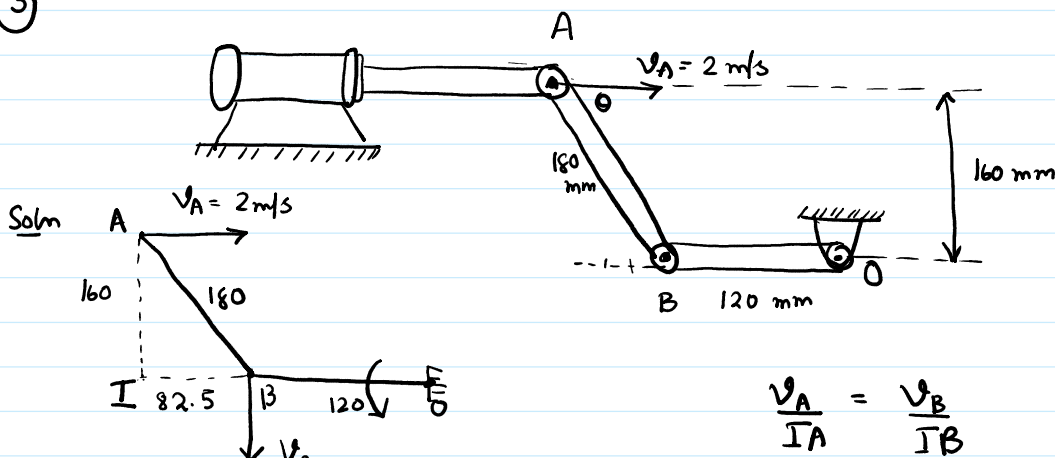
$$IG = 0.5 \text{ m above } G.$$

$$IA = \sqrt{0.5^2 + 0.3^2} = IB$$

$$\frac{v_A}{IA} = \frac{v_B}{IB} = \frac{v_G}{IG} = 4$$

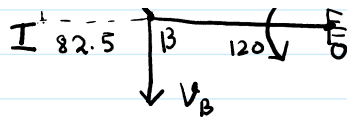
$$v_A = v_B = 2.33 \text{ m/s.}$$

③



Find out the angular velocity of OB by method of I-centre.

$$\frac{v_A}{IA} = \frac{v_B}{IB}$$



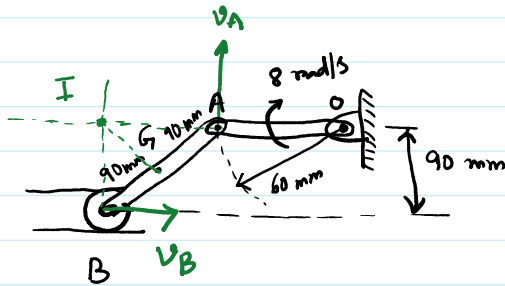
$$\frac{v_A}{I_A} = \frac{v_B}{I_B}$$

$$\frac{2}{0.16} = \frac{v_B}{0.0825} \Rightarrow v_B = 2 \times \frac{0.0825}{0.16} = 1.031 \text{ m/s}$$

$$\omega_{OB} = \frac{v_B}{OB} = 8.593 \text{ rad/s}$$

④

W



For the instant represented, when crank OA passes the horizontal position, determine velocity of center G of link AB by inst center method.

$$v_G = 277 \text{ mm/s}$$

$$\frac{v_A}{I_A} = \frac{v_B}{I_B} = \frac{v_G}{I_G}$$